

Jesús M. Rueda-Becerril | Ph.D.

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Professional Summary

Computational physicist with a PhD in Physics and ten years of experience building and running simulation codes for relativistic plasmas. Author of Paramo, an open-source Fortran library that solves the Fokker-Planck equation coupled to radiative transfer. Led an upgrade of the PatchworkMHD code on Frontera and used gradient-based optimization to calibrate model parameters against observations. Nine publications, three as first author, and Co-I on a funded NASA Fermi grant.

Skills

Simulation: Kinetic plasma modeling (Fokker-Planck), radiative processes (synchrotron, inverse Compton), radiative transfer, relativistic MHD, Lagrangian particle dynamics, numerical PDEs, finite difference/volume methods, numerical linear algebra, parameter estimation, uncertainty quantification

HPC & Parallel: OpenMP, OpenACC (GPU offloading), MPI, SLURM, PBS, HDF5

ML & Data: Scientific machine learning, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, Astropy, ML-assisted parameter estimation, gradient-based optimization, SVM, random forest, neural networks

Languages: C/C++, Python, Fortran 95, Rust, R, Java, SQL, Julia, Shell

Tools: Git, Docker, Jenkins, QGIS, Paraview, VisIt

Experience

Independent Contractor, Mercor.

Jan 2026 — Present

- Reviewed 50+ expert-level problems in physics, hydrodynamics, astrophysics, and cosmology; wrote original prompts that stumped frontier models.

Spatial Data Scientist, TealWaters.

May 2025 — Jan 2026

- Optimized the terrain-modeling code that computes elevation derivatives from 1 m, 3 m, and 10 m DEMs over 20 tiles, improving its computational performance.
- Mapped wetland probability across the 2,160 km² Skykomish watershed with the Wetland Intrinsic Potential (WIP) random-forest model, reaching >75% accuracy.

Software Engineer, Paychex.

Apr 2022 — Jan 2024

- Developed and tested production data-pipeline software (Java, Python, SQL) with CI/CD.

Independent Researcher & Open-Source Developer, Self-directed.

Jan 2024 — May 2025

- Co-developed Tleco, a Rust/Python toolkit that solves the Fokker-Planck equation coupled to radiative transfer for relativistic outflows (ApJ 2024).
- Developing WindsOfChange with the UW Forest Mycobiome Lab: a C++/R library for wind-driven spore dispersal over hilly forest terrain (RcppParallel, OpenMP).

Postdoctoral Research Associate, Rochester Institute of Technology.

Feb 2021 — Apr 2022

- Led a 7-person team in an NSF-sponsored upgrade of PatchworkMHD, a C code for MHD simulations of supermassive black hole binaries, adding black hole spin with under 5% runtime overhead.
- Ran and benchmarked large production simulations on Frontera (TACC), about 1,200 node-hours, and identified performance bottlenecks in the parallel code.
- Calibrated Paramo parameters against Fermi-LAT observations using gradient-based optimization, comparing simulation output with data and updating the model automatically; supervised a graduate student on the same problem.
- Collaborated on a multi-institutional HPC project on binary neutron star mergers, resulting in 2 publications.

Postdoctoral Research Fellow, Purdue University.

Oct 2018 – Nov 2020

- Designed and wrote Paramo, an open-source Fortran 95 library that solves the Fokker-Planck equation for relativistic particle distributions coupled to radiative transfer. OpenMP gives about 20× speedup on 5 cores; GPU offload tested with OpenACC.
- Implemented implicit PDE solvers, numerical integration, and arbitrary-shape particle distributions without slowing the simulation.
- Wrote Python tools to analyze Paramo output and run parameter optimization; other groups have adopted Paramo, including Combi & Siegel (2023).
- Co-I on two NASA Fermi Guest Investigator proposals, one funded (\$68K) to model blazar radiation; mentored 3 graduate students and published 2 peer-reviewed papers.

Postdoctoral Research Fellow, Universidad Michoacana de San Nicolás de Hidalgo.

Jan – Sep 2018

- Studied how magnetization of shocked plasma shapes the radiation spectra of relativistic internal shocks, using the C-SPEV simulation code.

Graduate Research Assistant, *Universitat de València*.

Oct 2011 – Jul 2017

- Independently developed Shell and Python pipelines to run the C-SPEV radiative transfer simulation code and process large HDF5 output datasets: 10–60 simulations per project and tens of GB of output.
- Implemented numerical tools and data handling extensions to C-SPEV for computing both discrete and continuous spectra from particle distributions with arbitrary shape.
- Published 2 first-author papers on numerical simulations of relativistic plasma dynamics.

Projects

Blazar Population ML, github.com/altjerue/ilwikak-ml.

Jan 2026 — Present

Scientific ML (Python, scikit-learn, PyTorch)

- Applying classical ML and Bayesian inference to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength observational data.
- Building a neural network to learn the mapping from physical jet parameters to observable SEDs accurately enough to compare with Tleco results.
- Forecasting blazar gamma-ray flares using a general-purpose time series foundation model, fine-tuned with Tleco-generated synthetic light curves.

WindsOfChange, github.com/UW-ForestMycobiomeLab/WindSporesHills.

Aug 2024 — Present

Open-Source Lagrangian Particle Simulation Library (R/C++)

- Developed C++ library implementing Lagrangian particle dispersion, atmospheric turbulence, and fluid flow over complex hilly terrain with forest canopy, based on analytical models (Massman & Weil 1999; Finnigan & Belcher 2004).
- Parallelized particle tracking using RcppParallel and OpenMP; designed for scalability to tens of thousands of particles across kilometer-scale domains.

Tleco, github.com/zkdavis/Tleco.

Jan 2024 — Sep 2024

Open-Source Physics Simulation Library

- Multiscale/multiphysics simulation toolkit combining Rust and Python interfaces for modeling relativistic particle distributions via Fokker-Planck equations coupled with radiative transfer.
- Implements numerical PDE solvers, particle distribution algorithms, and spectral calculation methods for relativistic plasma physics.

Paramo, github.com/altjerue/paramo.

Oct 2018 — Apr 2022

Open-Source Physics Simulation Code (CONAHCYT Funded)

- Modular physics simulation library (Fortran 95 + OpenMP) modeling relativistic particle radiation; predecessor to Tleco.

Selected Publications (9 total, 3 first-author)

- Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, *ApJ* **976**, 182, (2024), [arXiv:2405.17581](https://arxiv.org/abs/2405.17581).
- **Rueda-Becerril, J. M.**, Harrison, A. O. & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, *MNRAS* **501**, 4092–4102, (2021), [arXiv:2009.02273](https://arxiv.org/abs/2009.02273).
- **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, *MNRAS* **468**, 1169–1182, (2017), [arXiv:1612.06383](https://arxiv.org/abs/1612.06383).

Education

Universitat de València

Valencia, Spain

Ph.D. in Physics. Excellent Cum Laude.

Jul. 2017

Universidad Michoacana de San Nicolás de Hidalgo

Morelia, Mexico

M.Sc. in Physics.

Sep. 2011

Universidad Autónoma del Estado de México

Toluca, Mexico

B.Sc. in Physics.

Dec. 2008